

NexaTel Customer Churn Analytics

Data Quality Report — Phase 1: Data Cleaning & Validation

1. Dataset Overview

The NexaTel dataset contains 24 interconnected tables covering customers, subscriptions, billing, payments, usage, network quality, complaints, support, campaigns and geographical information. The raw datasets were loaded and profiled before cleaning. The original files in **data/raw/** were preserved without modification.

2. Data Quality Issues Identified

- Duplicate customer records and duplicate billing invoices.
- Duplicate payment records.
- Invalid customer ages, including negative and unrealistic values.
- Invalid email formats.
- Invalid phone numbers and PIN codes.
- Inconsistent capitalization in customer city names.
- Misspelled city names.
- Mixed date formats in customer acquisition dates.
- Future billing dates.
- Negative invoice amounts.
- Incorrect GST amounts.
- Broken payment-to-invoice foreign keys.
- Invalid subscription status values.
- Missing subscription plan references.
- Extreme/outlier values in data usage.
- Missing customer address, email and income values.

3. Cleaning Actions Performed

Issue / Area	Cleaning action
Duplicate customers	Removed 76 duplicate customer records.
Duplicate billing invoices	Removed 344 duplicate invoice records.
Duplicate payments	Removed 276 duplicate payment records.
Customer age	Corrected invalid ages using date-of-birth-based validation.
Email	Converted invalid email values to missing values.
Phone / PIN	Converted invalid phone numbers and PIN codes to missing values.
City capitalization	Standardized customer city capitalization.
Misspelled cities	Corrected 380 misspelled city records using the official city reference table.
Acquisition dates	Standardized mixed formats to DD-MM-YYYY.
Billing dates	Converted 229 future billing dates to missing values.
Invoice totals	Converted 229 negative invoice totals to missing values.

Issue / Area	Cleaning action
GST	Recalculated incorrect GST values using the expected 18% GST relationship.
Payment invoice FK	Converted 184 broken invoice references to missing values.
Subscription status	Converted 102 invalid UNKNOWN_STATE values to missing values.
Subscription plan FK	Retained 41 missing plan references as missing.
Data usage	Converted 171 extreme usage values above 350 GB to missing values.

4. Missing-Value Assessment

Remaining missing values were reviewed according to business context rather than being automatically imputed.

- Missing customers.churn_date values correspond to active customers.
- Missing subscriptions.end_date values primarily correspond to active subscriptions.
- Missing support-ticket resolution fields correspond to unresolved tickets.
- Missing complaint resolution dates correspond to unresolved complaints.
- Missing customer income, address and email values were retained because their correct values could not be reliably determined from the available data.

5. Referential Integrity

Foreign-key validation was performed across the relevant interconnected tables. All tested foreign-key relationships returned zero unresolved invalid references after cleaning.

6. Final Validation

Validation	Final result
Tables retained	24 / 24
Duplicate rows across all 24 tables	0
Duplicate primary keys in customers, billing and payments	0
Invalid ages	0
Invalid email formats	0
Invalid phone numbers	0
Invalid PIN codes	0
Future billing dates	0
Negative invoice amounts	0
Invalid subscription statuses	0
Extreme usage values	0

7. Data Quality Log Comparison

The data_quality_issue_log was used as a reference for intentionally introduced data-quality issues. In several cases, detected row counts differed slightly from the log because duplicate records caused multiple row occurrences for the same affected entity. A few missing-value counts also differed from the log and were retained according to the actual values present in the raw datasets rather than being artificially changed to match the log.

8. Output and Conclusion

The NexaTel datasets have been cleaned, validated and saved as 24 cleaned CSV files in **data/cleaned/**. The raw datasets remain unchanged in **data/raw/**. The cleaned datasets are ready for exploratory data analysis, visualization and churn KPI calculation.

Phase 1 Status: Complete — Data Cleaning & Quality Validation